



TARGET CAT 2025



MBA FASTRACK

QUANT : Arithmetic

Time Speed Distance

Part I

Lecture No.- 07

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Recap of Previous Lecture :

$$\left. \begin{array}{l} A \rightarrow 10 \\ B \rightarrow 15 \\ C \rightarrow 20 \end{array} \right\}$$

Wages
Alternate
together
effi

TOPICS *to be covered*



- 1) ~~Concept~~ & Numerical based on 'Time & Work (When Workers are in LOT)'
- 2) Numerical based on 'Basic Concepts of Distance Time & Speed'
- 3) Concept based on 'Relative Speed'
- 4) Problems for Practice & CAT PYQs



Work Rate & Time : [When Workers are in LOT]



Ex

$$T \cdot w \cdot D = m \cdot d$$

①

$$m = 60$$

$$d = 20$$

$$m = ?$$

$$d = 24$$

$$\begin{aligned} T \cdot w \cdot D &= 60 \times 20 \\ &= 1200 \end{aligned}$$

$$\begin{aligned} T \cdot w \cdot D &= m \cdot d \\ 1200 &= m \cdot (24) \end{aligned}$$

$$50 = m$$

Ex

$$T \cdot w \cdot D = m \cdot d \cdot h$$

$$m = 60$$

$$d = 20$$

$$h = 8$$

$$m = 20$$

$$d = ?$$

$$h = 10$$

$$\begin{aligned} T \cdot w \cdot D &= (60)(20)(8) \\ &= 9600 \end{aligned}$$

$$T \cdot w \cdot D = m \cdot d \cdot h$$

$$9600 = 20(d)(10)$$

$$\frac{9600}{200} = d$$

$$48 = d$$

$$T \cdot w \cdot D = w \cdot d$$

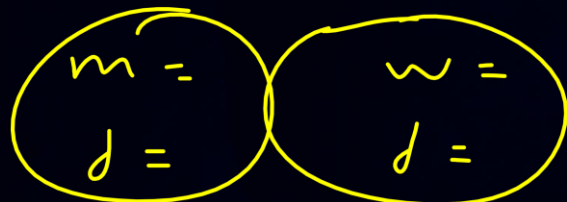
$$T \cdot w \cdot D = m \cdot h$$

$$T \cdot w \cdot D = b \cdot h$$

$$T \cdot w \cdot D = g \cdot h$$



Work Rate & Time : [When Workers are in LOT]



Same Question

QUESTION-1



#Q. If 6 men and 8 boys can do a piece of work in 10 days while 26 men and 48 boys can do the same in 2 days, the time taken by 15 men and 20 boys in doing the same type of work will be:

$$6m \& 8b @ 10 \text{ days}$$

$$T \cdot W \cdot D = 60md + 80bd$$

$$26m \& 48b @ 2 \text{ days}$$

$$T \cdot W \cdot D = 52md + 96bd$$

A 4 days

B 5 days

C 6 days

D 7 days

$$T \cdot W \cdot D = 60(2) + 80(1) = 200$$

$$60md + 80bd = 52md + 96bd$$

$$8md = 16bd$$

$$\frac{md}{bd} = \frac{16}{8} = \frac{2}{1}$$

Let, $md \rightarrow 2$
 $bd \rightarrow 1$

$$15m \& 20b @ x \text{ days}$$

$$T \cdot W \cdot D = 15x(md) + 20x(bd)$$

$$200 = 15x(2) + 20x(1)$$

$$200 = 50x$$

$$4 = x$$

QUESTION- 2



#Q. Humans and robots can both perform a job but at different efficiencies. Fifteen humans and five robots working together take thirty days to finish the job, whereas five humans and fifteen robots working together take sixty days to finish it. How many days will fifteen humans working together (without any robot) take to finish it?

15 h & 5 r @ 30d | 5 h & 15 r @ 60 days [CAT 2018: Slot 1]

$$T \cdot w \cdot D = 450 h d + 150 r d \quad | \quad T \cdot w \cdot D = 300 h d + 900 r d$$

A 40

B 32

C 36

D 45

$$\begin{aligned}
 T \cdot w \cdot D &= 450(15) + 150(5) \\
 &= 2250 + 750 \\
 &= 3000
 \end{aligned}$$

$$\begin{aligned}
 150(h d) &= 750(r d) \\
 \frac{h d}{r d} &= \frac{750}{150} = \frac{5}{1} \\
 \text{let, } h d &= 5 \\
 r d &= 1
 \end{aligned}$$

15 h @ x days

$$\begin{aligned}
 T \cdot w \cdot D &= 15x(h d) \\
 3000 &= 15x(5) \\
 \frac{3000}{75} &= x \\
 32 &= x
 \end{aligned}$$

QUESTION- 3

$$T \cdot w \cdot D = m \cdot l \cdot h$$



#Q. A group of N people worked on a project. They finished 35% of the project by working 7 hours a day for 10 days. Thereafter, 10 people left the group and the remaining people finished the rest of the project in 14 days by working 10 hours a day. Then the value of N is

[CAT 2022: Slot 3]

A 23

B 140

C 36

D 150

$$\begin{array}{l} m \rightarrow N \\ h \rightarrow 7 \\ d \rightarrow 10 \end{array}$$

$$\begin{array}{l} m \rightarrow (N-10) \\ h \rightarrow 10 \\ d \rightarrow 14 \end{array}$$

$$N \cdot 7 \cdot 10 = \frac{35}{100} \cdot T \cdot w \cdot D \quad (N-10) \cdot 10 \cdot 14 = \frac{65}{100} T \cdot w \cdot D$$

$$\frac{N \cdot 7 \cdot 10}{(N-10) \cdot 10 \cdot 14} = \frac{\frac{35}{100} \cdot T \cdot w \cdot D}{\frac{65}{100} \cdot T \cdot w \cdot D}$$

$$\frac{N}{(2N-20)} = \frac{35}{65}$$

$$14N - 140 = 13N$$

$$N = 140$$



Time Speed Distance

Time Speed Distance:

$$1 \frac{\text{km}}{\text{h}} = \frac{1000 \text{ m}}{3600 \text{ sec}} = \frac{5}{18} \text{ m/s}$$

$$\text{Speed} = \frac{\text{Dist}}{\text{Time}}$$

$$\text{Time} = \frac{\text{Dist}}{\text{Speed}}$$

$$\text{Speed} \times \text{Time} = \text{Dist}$$

$d \rightarrow \text{km}$	$d \rightarrow \text{m}$
$t \rightarrow \text{hrs}$	$t \rightarrow \text{sec}$
$s \rightarrow \text{km/h}$	$s \rightarrow \text{m/s}$

$$\textcircled{1} \quad 1 \text{ km/h} = \frac{5}{18} \text{ m/s}$$

$$\text{e.g. } 72 \text{ km/h} = 72 \times \frac{5}{18} \text{ m/s} = 15 \text{ m/s}$$

$$\textcircled{2} \quad 1 \text{ m/s} = \frac{18}{5} \text{ km/h}$$

$$\text{e.g. } 20 \text{ m/s} = 20 \times \frac{18}{5} \text{ km/h} = 72 \text{ km/h}$$

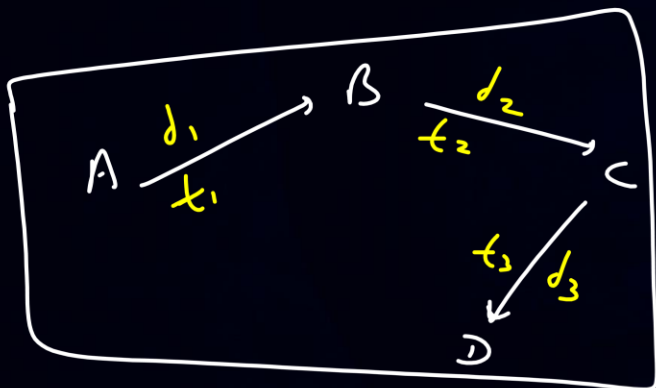
$$\textcircled{3} \quad 1 \text{ min} = \frac{1}{60} \text{ hr}$$



Time Speed Distance :

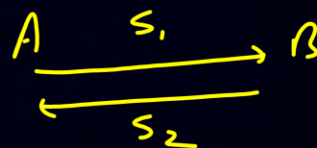


Average Speed :



$$\begin{aligned}\text{Avg. Speed} &= \frac{\text{T. Dist}}{\text{T. Time}} \\ &= \frac{d_1 + d_2 + d_3}{t_1 + t_2 + t_3}\end{aligned}$$

Special Case: [dist. is same with both speeds]



$$\text{Avg. Speed} = \frac{2s_1 s_2}{s_1 + s_2}$$

a & b

$$H.M = \frac{2}{\left(\frac{1}{a} + \frac{1}{b}\right)} = \frac{2}{\left(\frac{a+b}{ab}\right)}$$

$$= \frac{2ab}{a+b}$$

a, b, c

$$H.M = \frac{3}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}$$

a, b, c, d

$$H.M = \frac{4}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}}$$



Time Speed Distance :



Proportionality:

(I)

dist. is constant

Direct

$A \uparrow B \uparrow$
 $A \downarrow B \downarrow$

Inverse

$A \uparrow B \downarrow$
 $A \downarrow B \uparrow$

$$S \propto \frac{1}{t}$$

$$\frac{S_1}{S_2} = \frac{t_2}{t_1}$$

(I)

$$\begin{array}{l} S_1 = 50 \text{ km/h} \\ t_1 = 3 \text{ h} \end{array}$$

Same dist
 $A \longleftrightarrow B$

(II)

$$\begin{array}{l} S_2 = ? \\ t_2 = 30 \text{ h} \end{array}$$

$$\frac{S_1}{S_2} = \frac{t_2}{t_1}$$

$$\frac{50}{S_2} = \frac{30}{3}$$

$$10(S_2) = 50$$

$$S_2 = 5 \text{ km/h}$$



Time Speed Distance :



II

Time is constant

$$S \propto d$$

$$\frac{S_1}{S_2} = \frac{d_1}{d_2}$$

$$\Rightarrow \frac{10}{2.5} = \frac{S}{d_2}$$

$\angle (d_2) = 5$

$$d_2 = 1.25 \text{ km}$$

$$\begin{array}{l|l} S_1 = 10 \text{ km/h} & S_2 = 2.5 \text{ km/h} \\ d_1 = 5 \text{ km} & d_2 = ? \end{array}$$



Time Speed Distance :



III

Speed is constant

$$d \propto t$$

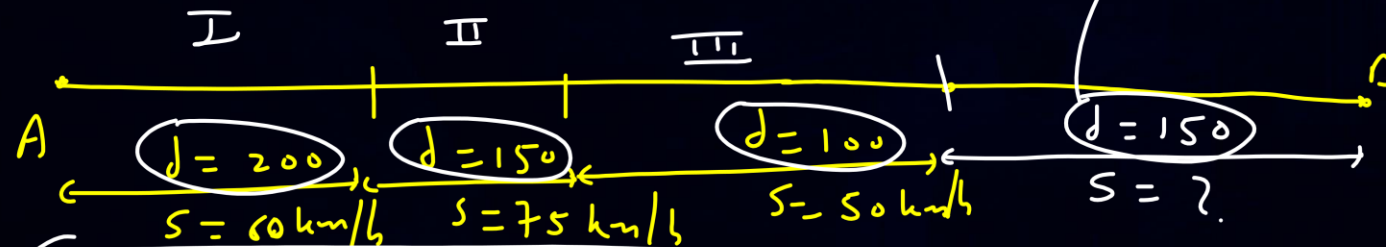
$$\frac{d_1}{d_2} = \frac{t_1}{t_2}$$

QUESTION- 4



#Q. A truck covers a distance of 600 km in 9 hours. If it covers the first 200 km at 60 km/h, the next 150 km at 75 km/h, and the next 100 km at 50 km/h, at what constant speed must it travel the last leg of the journey?

90 T. dist = 600 km



$t_1 = \frac{d}{s} = \frac{200}{60} = \frac{10}{3}$	$t_2 = \frac{150}{75} = 2$	$t_3 = \frac{100}{50} = 2$
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$$t_4 = 9 - \left[\frac{10}{3} + 2 + 2 \right]$$

$$= 9 - \left[\frac{10}{3} + 4 \right]$$

$$= 9 - \frac{22}{3} = \frac{5}{3} \text{ hrs}$$

T. time = 9 hrs

$$\therefore S = \frac{d}{t} = \frac{150}{(5/3)}$$

$$= 150 \times \frac{3}{5}$$

$$= 90 \text{ km/h}$$

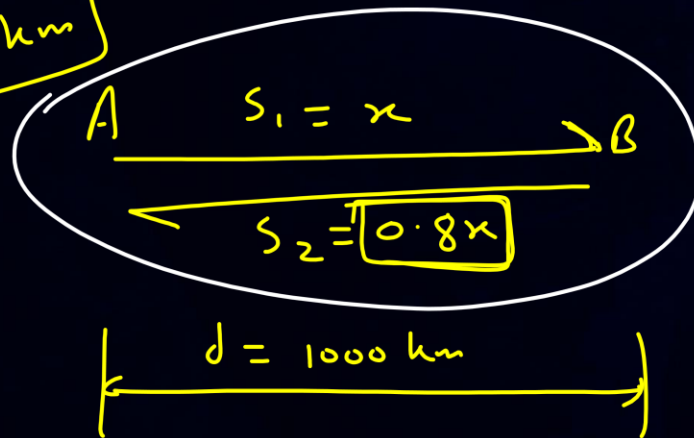
QUESTION- 5

$$t = \frac{d}{s}$$



#Q. The average speed of a train is 20% less on the return journey than on the onward journey. (The train halts for an hour at the destination station) before starting on the return journey. If the total time taken for the to-and-fro journey is 46 hrs., covering a distance of 2000 km, the speed of the train on the return journey is:

T. dist = 2000 km



A 50 kmph

B 60 kmph

C ☒ 40 kmph

D 65 kmph

$$\text{T. time} = 46 \text{ hrs}$$

$$\therefore \text{T. travelling time} = 46 - 1 = 45$$

$$t_1 + t_2 = 45$$

$$\frac{1000}{x} + \frac{1000}{0.8x} = 45$$

$$\frac{800}{0.8x} + \frac{1000}{0.8x} = 45$$

$$\frac{1800}{0.8x} = 45$$

$$\frac{1800}{45} = 0.8x$$

$$40 = 0.8x$$

QUESTION- 6



#Q. A helicopter flies along the sides of a square field of side length 100 kms. The first side is covered at a speed of 100 kmph, and for each subsequent side the speed is increased by 100 kmph till it covers all the sides. The average speed of the helicopter is

A 184 kmph $S_4 = 400$

B 200 kmph $d = \frac{100}{400}$

C 250 kmph

D 192 kmph

$S_1 = 100 \text{ km/h}$ $t_1 = \frac{d_1}{S_1} = \frac{100}{100} = 1$
 $d_1 = 100 \text{ km}$
 $d_2 = 100 \text{ km}$
 $S_2 = 200 \text{ km/h}$
 $t = \frac{100}{200} = \frac{1}{2}$
 $d_3 = 100$
 $S_3 = 300 \text{ km/h}$
 $t = \frac{100}{300} = \frac{1}{3}$
 $d_4 = 100$
 $S_4 = 400$
 $d = \frac{100}{400}$
 $d = \frac{1}{4}$

$$\text{Avg Speed} = \frac{T \cdot \text{dist}}{T \cdot \text{time}}$$

$$= \frac{400}{\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}}$$

$$= \frac{400}{\left(\frac{12+6+4+3}{12}\right)}$$

$$= \frac{400}{\frac{25}{12}} = \frac{4800}{25} = 192$$

$$\text{Avg. Speed} = \frac{4}{\frac{1}{S_1} + \frac{1}{S_2} + \frac{1}{S_3} + \frac{1}{S_4}}$$

$$= \frac{4}{\left(\frac{1}{100}\right) + \left(\frac{1}{200}\right) + \frac{1}{300} + \frac{1}{400}}$$

$$= \frac{4}{\left[\frac{12}{1200} + \frac{6}{1200} + \left(\frac{4}{1200}\right) + \left(\frac{3}{1200}\right)\right]}$$

$$= \frac{4}{\left(\frac{25}{1200}\right)}$$

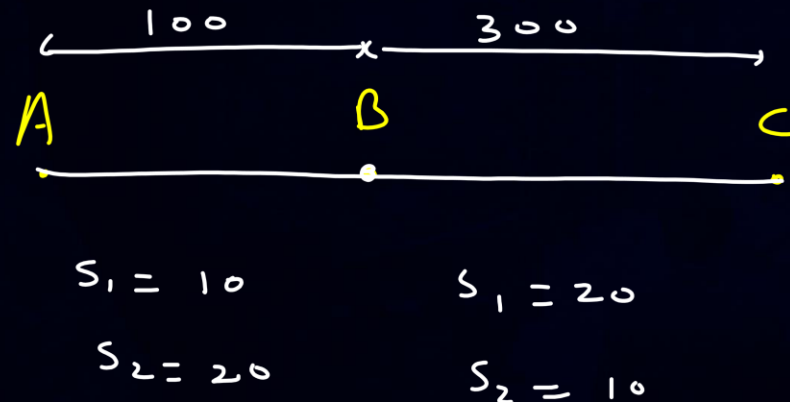
$$= \frac{4 \times 1200}{25} = 192$$

QUESTION- 7

$$t = \frac{d}{s}$$



#Q. The distance from B to C is thrice that from A to B. Two trains travel from A to C via B. The speed of train 2 is double that of train 1 while traveling from A to B and their speeds are interchanged while traveling from B to C. The ratio of the time taken by train 1 to that taken by train 2 in travelling from A to C is **[CAT 2020 : Slot 2]**



$$\frac{t_1}{t_2} = \frac{\left[\frac{100}{10} + \frac{300}{20} \right]}{\left[\frac{100}{20} + \frac{300}{10} \right]}$$

$$= \frac{10 + 15}{5 + 30} = \frac{25}{35} = \frac{5}{7}$$

A 7:5 Train I

B 4:1 Train II

C 1:4

D 5:7

QUESTION- 8

$$d = s \times t$$



#Q. When Geeta increases her speed from 12 km/hr to 20 km/hr, she takes one hour less than the usual time to cover the distance between her home and office. The distance between her home and office is 30 km

(I)

$$S_1 = 12 \text{ km/h}$$

$$t_1 = x \text{ hrs}$$

(II)

$$S_2 = 20 \text{ km/h}$$

$$t_2 = (x-1) \text{ hrs}$$



$$\therefore d = S_1 \times t_1$$

$$= 12 \times 2.5$$

$$= 30 \text{ km}$$

$$S \propto \frac{1}{t}$$

$$\frac{S_1}{S_2} = \frac{t_2}{t_1}$$

$$\frac{12}{20} = \frac{(x-1)}{x}$$

$$5x - 5 = 3x$$

$$2x = 5$$

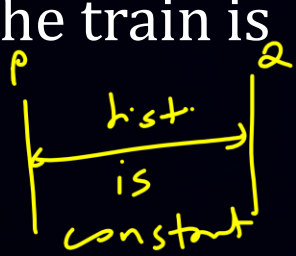
$$x = 2.5$$

QUESTION- 9

$$d = s \times t$$



#Q. A train travelled a certain distance at a uniform speed. Had the speed been 6 km per hour more, it would have needed 4 hours less. Had the speed been 6 km per hour less, it would have needed 6 hours more. The distance, in km, travelled by the train is [CAT 2024 : Slot 3]



original
 $S_1 \rightarrow x \text{ km/h}$
 $t_1 \rightarrow y \text{ hrs}$

(I)

$$S_1 = (x + 6) \text{ km/h}$$

$$t_2 = (y - 4) \text{ hrs}$$

(III)

$$S_3 = (x - 6) \text{ km/h}$$

$$t_3 = (y + 6) \text{ hrs}$$

A

780

$$d = xy$$

$$= 30 \times 26$$

$$= 780$$

$$d_1 = d_2 = d_3$$

$$xy = (x + 6)(y - 4) = (x - 6)(y + 6)$$

B

720

$$xy = xy - 4x + 6y - 24$$

$$4x = 6y - 24$$

C

640

$$4x = 6y - 24$$

$$2x = 3y - 12$$

$$2x = 3(x - 6) - 12$$

$$2x = 3x - 18 - 12$$

D

800

$$30 = 4$$

$$xy = xy + 6x - 6y - 36$$

$$6y = 6x - 36$$

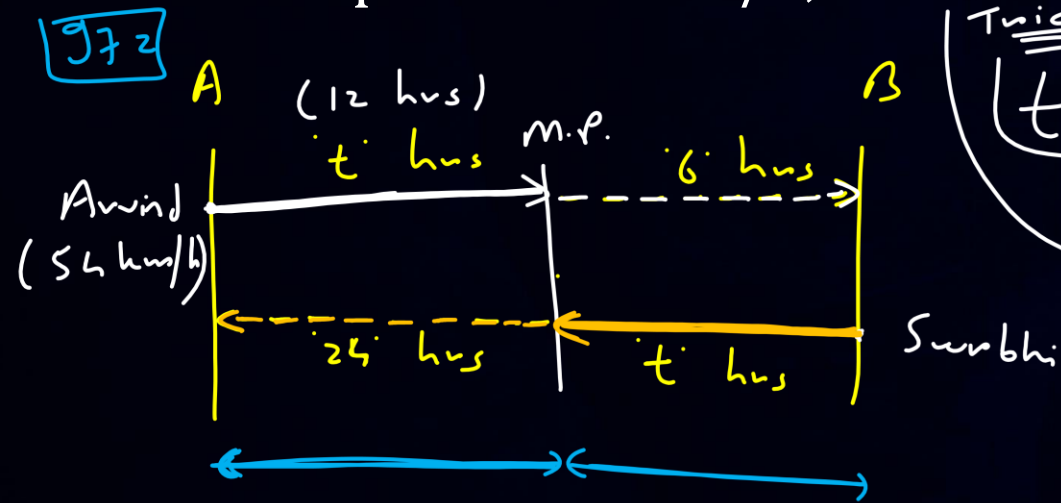
$$y = x - 6$$

$$y = 26$$

QUESTION- 10



#Q. Arvind travels from town A to town B, and Surbhi from town B to town A, both starting at the same time along the same route. After meeting each other, Arvind takes 6 hours to reach town B while Surbhi takes 24 hours to reach town A. If Arvind travelled at a speed of 54 km/h, then the distance, in km, between town A and town B is 972



Trick

$$t = \sqrt{T_1 \cdot T_2} = \sqrt{6 \times 24} = \sqrt{144} = 12$$

where $T_1 \rightarrow$ time by Ist person from m.p. to Destination

$T_2 \rightarrow$ time by IInd person from m.p. to Destination

Arvind Vs Surbhi

A to m.p.
m.p. to B

$t \rightarrow 24$
 $6 \rightarrow t$

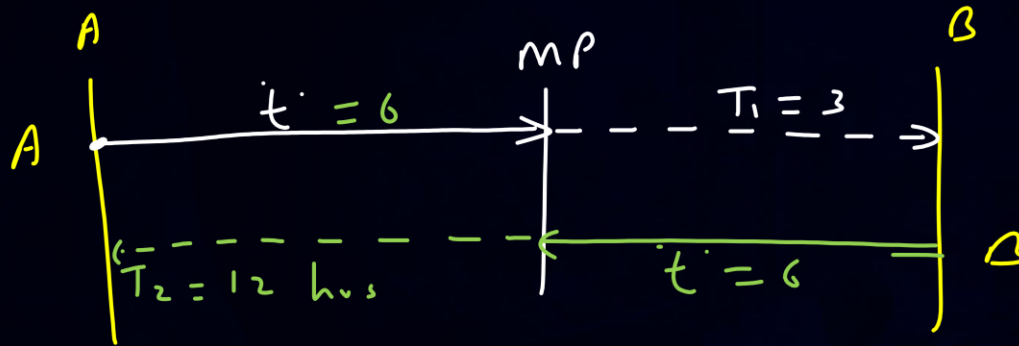
$t^2 = 144$
 $t = 12$

Arvind
 $S = 54 \text{ km/h}$
time $= 12 + 6 = 18 \text{ h}$
 $d = S \times t = 54 \times 18$
 $= \underline{\underline{972 \text{ km}}}$

QUESTION- 11



#Q. Two persons start from points A and B toward each other at the same time. After crossing, A takes 3 hours to reach B, and B takes 12 hours to reach A. A bird flies back and forth between them at 40 km/h. How far does the bird fly before they meet?



$$t = \sqrt{T_1 \cdot T_2}$$
$$= \sqrt{3 \times 12} = \sqrt{36} = 6$$

Bird

$$S = 40 \text{ km/h}$$

$$t = \boxed{6 \text{ hrs}}$$

$$d = S \times t$$

$$= 40 \times 6 = 240 \text{ km}$$

A 200 km

B 240 km

C 120 km

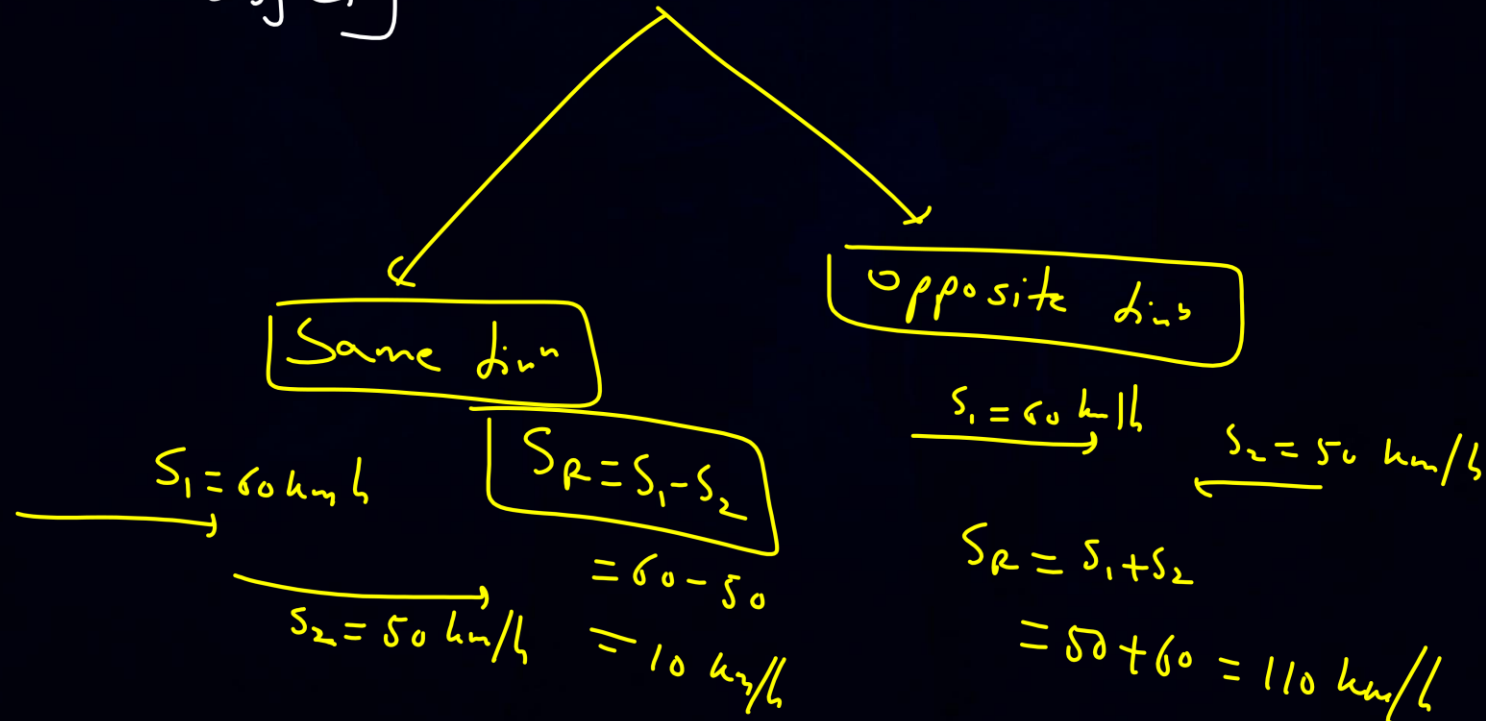
D 150 km



Relative Speed :



[Speed of one moving object
with respect to other moving
object]





Relative Speed :



$$s_1 > s_2$$

$$s_R = s_1 - s_2 \longrightarrow \text{Same dir}$$

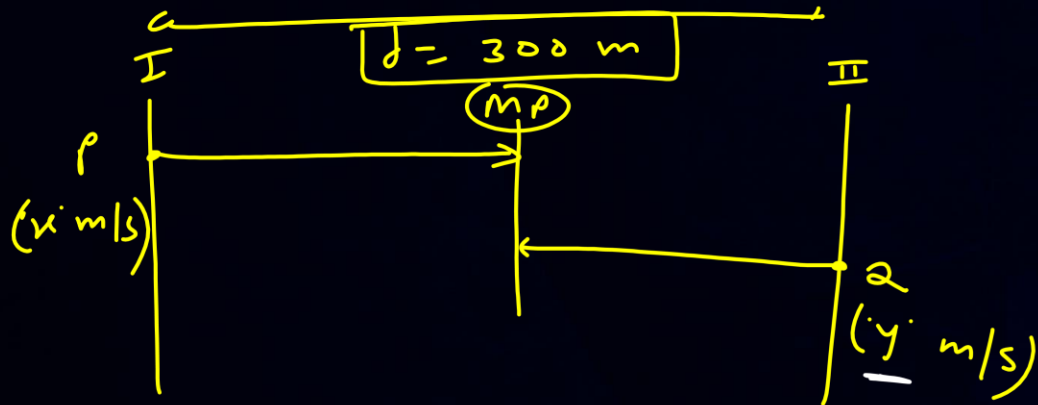
$$s_R = s_1 + s_2 \longrightarrow \text{oppo. dir}$$

QUESTION- 12

$$150\% \text{ of } x = 1.5x \quad | \quad 75\% \text{ of } y = 0.75y$$



#Q. Two people P and Q start simultaneously from opposite ends of a 300-meter track and run toward each other. They meet after 10 seconds. If P increases his speed by 50% and Q decreases his speed by 25%, they meet after 8 seconds. What is Q's original speed (in m/s)? 10 original



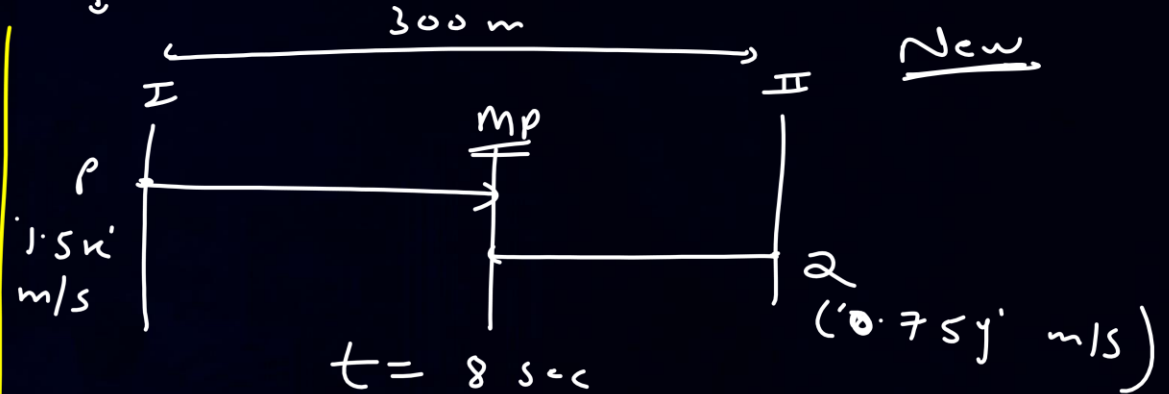
$$t = 10 \text{ sec}$$

$$\text{dist} = 300 \text{ m}$$

$$S_r = x + y$$

$$S_r = \frac{\text{dist}}{\text{time}} \Rightarrow x + y = \frac{300}{10} = 30$$

$$x = 30 - y$$



$$t = 8 \text{ sec}$$

$$\text{dist} = 300 \text{ m}$$

$$S_r = 1.5x + 0.75y$$

$$S_r = \frac{\text{dist}}{\text{time}}$$

$$1.5x + 0.75y = \frac{300}{8} = 37.5$$

$$150x + 75y = 3750$$

$$2x + y = 50$$

$$2(30 - y) + y = 50$$

$$60 - 2y + y = 50$$

$$10 = y$$

QUESTION- 13



#Q. A and B start from two points 60 km apart and walk toward each other. A walks at 4 km/h, and B walks at 6 km/h. A dog runs from A to B and back continuously at 12 km/h. Find the distance travelled by the dog, by the time A and B meet?



$$\begin{aligned} \text{dist} &= 60 \\ S_R &= 4 + 6 = 10 \text{ km/h} \\ t &= \frac{60}{S_R} = \frac{60}{10} = 6 \text{ h} \end{aligned}$$

$$\begin{aligned} \underline{\text{Dog}} \\ S &= 12 \text{ km/h} \\ \boxed{t} &= 6 \\ d &= ? \end{aligned}$$

$$\begin{aligned} d &= S \times t \\ &= 12 \times 6 \\ &= 72 \end{aligned}$$

A 72 km

B 60 km

C 48 km

D 36 km

QUESTION- 14



#Q. Car A leaves Delhi at a certain time, after 5 hours Car B leaves Delhi in the same direction as of A. Speed of Car A is 20 km/hr and Speed of Car B is 60 km/hr. How much time Car B will take to overtake Car A?

TRY
(Next Class)

- A** 2.5 hours
- B** 3.5 hours
- C** 3 hours
- D** 2 hours

QUESTION- 15



#Q. Ravi is driving at a speed of 40 km/h on a road. Vijay is 54 meters behind Ravi and driving in the same direction as Ravi. Ashok is driving along the same road from the opposite direction at a speed of 50 km/h and is 225 meters away from Ravi. The speed, in km/h, at which Vijay should drive so that all the three cross each other at the same time, is

[CAT 2023 : Slot 2]

TRY
(Next Class)

A 58.8

B 64.4

C 67.2

D 61.6



2 Mins Summary

- 1) ~~Concept~~ & Numerical based on 'Time & Work (When Workers are in LOT)'
- 2) ~~Numerical~~ based on 'Basic Concepts of Distance Time & Speed'
- 3) ~~Concept~~ based on 'Relative Speed'
- 4) ~~Problems for Practice~~ & CAT PYQs



*Thank
You*